1. Explain the difference between an event-driven application and a console-based application

   An event-driven application executes code in response to events. This means that the GUI or application will do nothing until and unless the user reacts- could be clicking a button or pressing a key- in which case an immediate response will take place.

   In a console based application, however, it follows a series of steps that has a start and finish.

2. Explain how code is executed in an event-driven application

   An EDA executes code in response to events rather than following step-by-step sequences. The application consistently listens for an action and instantly responds by executing a specific method called an event handler. Swing components also use listeners to determine if an event or any change has occurred.

3. Can components be added directly to a frame? Explain.

   Yes, other components can be directly added to a frame, however, an invisible inner container called a JPanel is usually needed. This allows us control over the layout and stops components from overlapping and hiding behind one another.

4. Can a label respond to events? Explain.

   Generically, a label is supposed to display text that cannot be changed whatsoever by the user. It can however be programmed to respond to different events, display different texts or icons to the user.

5. Why do you think a GUI needs to be run from an event-dispatching thread?

   GUIs need to be run from an event-displaching thread to ensure that each event handler finishes executing before the next one executes. This prevents overloading the program while making sure program components operate correctly.

6. What is the difference between a label and a button?

   A label is meant as a display screen that cannot be interacted with by the user, while a JButton can respond to the interaction from the user.

8. List three ways to control the layout of a content pane.

In a Swing GUI, the layout of a content pane can be controlled by adding borders, using a layout manager, and setting alignments.

13. What must first be done with numeric data typed in a text field before it can be used in a calculation?

It must be converted ( using the method parsed) from a String into a numeric data type like int or double before it can be used in a calculation.

14. What is the value of num1 in the last statement below?

The value of num1 = 8.0

double num1; ( establishes num 1 as double)
Double num2 = new Double(3); ( num2 variable stores numerical 3)
String num3 = "5"; ( num3 variable stores “5”)
num1 = num2.doubleValue() + Double.valueOf(num3).doubleValue(); (converts both to double, adds num 2 and num3 producing num1 )

15. An application prompts a user to select a name from a list of six names. Which is a better component choice: a text field or a combo box? Explain.

The better option would be the combo box, because the combo box makes it easier for the user to just select between the options you want to give it, rather than having them have to type out their response which can lead to potential issues.